

Proposal: Extending the University Guidance System for Multi-Department Operation

Prepared for: Guidance Department Leadership / Project Stakeholders

Status: For Review (Revision 3 — Re-architected for Isolated Service Model)

1. Problem/Need Identification

The current workspace consists of six isolated ASP.NET Core (net10.0) web applications running on separate high localhost ports (52141–52146) for individual campus departments (Library, Registrar, Finance, Faculty, Student Portal, Guidance). Each department app operates as an isolated silo with no mechanism for inter-service communication or cross-app trust. Because each department runs as an independent web application with its own isolated store, cross-department operations cannot be solved by simply adding a DepartmentId column to a single database schema. Without an explicit inter-service protocol:

- Staff cannot forward or refer student requests to another department application.
- Cross-department referrals would require manual, out-of-band communication (email/paper), breaking audit trails and ownership enforcement guarantees.
- Copying data manually between applications creates severe risks of PII leakage and data desynchronization.

2. Introduction

The original platform was designed as an isolated web application for the Guidance Department. Core capabilities—including Authentication, Request Management, Triage, Audit Logging, PII Masking, CSV Import, and Polly Notifications—originated in GuidanceDepartmentMain and are copied to target projects via the Install-GuidanceServices.ps1 installer script.

While this model guarantees physical runtime, process, and data store isolation per department, it lacks an Inter-Service Referral Protocol. This proposal defines the minimum set of changes required to enable cross-department operation and inter-service collaboration while respecting project isolation, student privacy, workflow integrity, and local project boundaries.

3. Project Objectives

1. **Establish Cross-App Trust:** Standardize machine-to-machine (M2M) authentication across ports 52141–52146 without breaking local session handling or role-based access control.
2. **Implement Inter-Service Referral Protocol:** Extend the request lifecycle to support multi-department referrals via resilient HTTP calls while enforcing data minimization.
3. **Preserve Departmental Isolation:** Maintain independent runtime, process, and data store boundaries without introducing monolithic database schemas or shared stores.

4. **Ensure Audit & Privacy Compliance:** Retain immutable audit logs for outbound/inbound referrals and enforce strict privacy rules to ensure confidential counseling notes remain within GuidanceDepartmentMain.
5. **Standardize Service Synchronization:** Use GuidanceDepartmentMain as the canonical reference source and propagate contracts/services via Install-GuidanceServices.ps1.

4. Proposed System / Solution

4.1 Department Identity & Cross-App Trust Architecture

- **Local Department Identity:** Each department's appsettings.json specifies its code and local endpoints (e.g., DepartmentCode: "Guidance", DepartmentCode: "Finance").
- **Service-to-Service Trust:** Standardize machine-to-machine (M2M) authentication (via shared/asymmetric JWT signing keys or inter-service API keys) for secure HTTP communication across ports 52141–52146.
- **Service Layer Propagation:** All inter-service contracts (Contracts/ServiceContracts.cs) and referral transport services originate in GuidanceDepartmentMain and are synchronized across projects using Install-GuidanceServices.ps1.

4.2 Cross-Department Referral Protocol

- **Lifecycle Extension:** Extend the state machine from Requested $\rightarrow$ Resolved to include ReferredOut (originating department) and ReferredIn / Requested (receiving department).
- **Inter-Service REST Endpoint:** Implement an inbound endpoint across all department projects:

POST /api/v1/incoming-referral

- **Data Minimization Safety Valve:** Referral payloads are restricted to StudentId, sanitized ReferralReason, originating reference ID, and department code. Full case notes and local history remain private to the originating department.
- **Resilient Outbound Delivery:** Outbound referral payloads pass through Polly retry and circuit-breaker pipelines in NotificationService. If the target department is offline or restarting, retries are handled safely.
- **Audit Trail Capture:** Both ReferralSent and ReferralReceived events are appended to each department's respective local AuditLogService.

5. Business Process Model

5.1 Department Actors & Roles (Swimlanes)

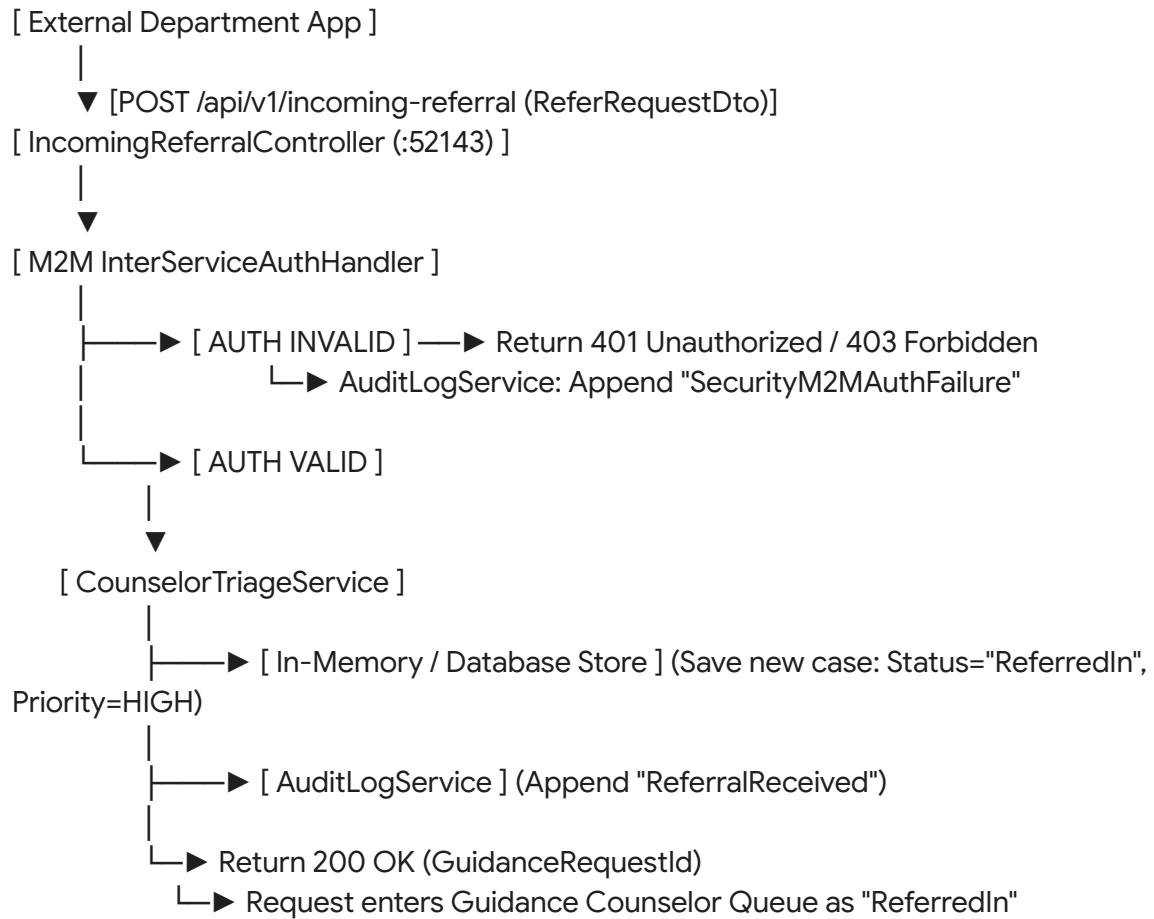
Actor / Component	Role	Boundary & Access
Student	Guidance client seeking support	Accesses request.html via claims-derived identity

		(Student role).
Guidance Counselor / Triage Officer	Case handler & triage staff	Accesses triage.html (Counselor role) to review queues, record confidential case notes, and resolve/refer cases.
Guidance Department Administrator	System lead / Department head	Accesses admin.html for local staff routing, endpoint config, and roster imports (Admin role).
Guidance Core Engine (GuidanceDepartmentMain)	Local service & business rule enforcement	Port 52143. Handles JWT verification, sanitization, in-memory/EF persistence, and PII masking.
External Department Endpoints	Receiving/Referring departments	Physical ports (52141–52146). Communicates via machine-to-machine (M2M) authenticated REST payloads.

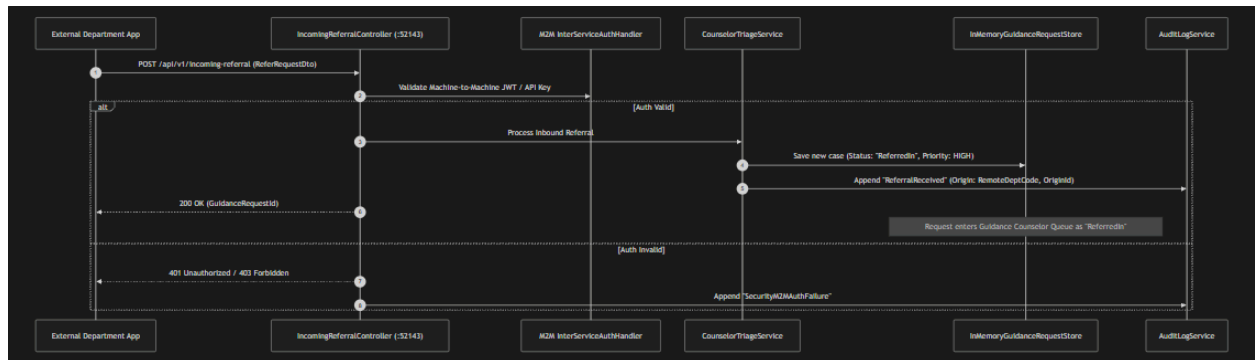
5.2 Outbound Referral Process Workflow

Actor / Component	Role	Boundary & Access
Student	Guidance client seeking support	Accesses <code>request.html</code> via claims-derived identity (<code>Student</code> role).
Guidance Counselor / Triage Officer	Case handler & triage staff	Accesses <code>triage.html</code> (<code>Counselor</code> role) to review queues, record confidential case notes, and resolve/refer cases.
Guidance Department Administrator	System lead / Department head	Accesses <code>admin.html</code> for local staff routing, endpoint config, and roster imports (<code>Admin</code> role).
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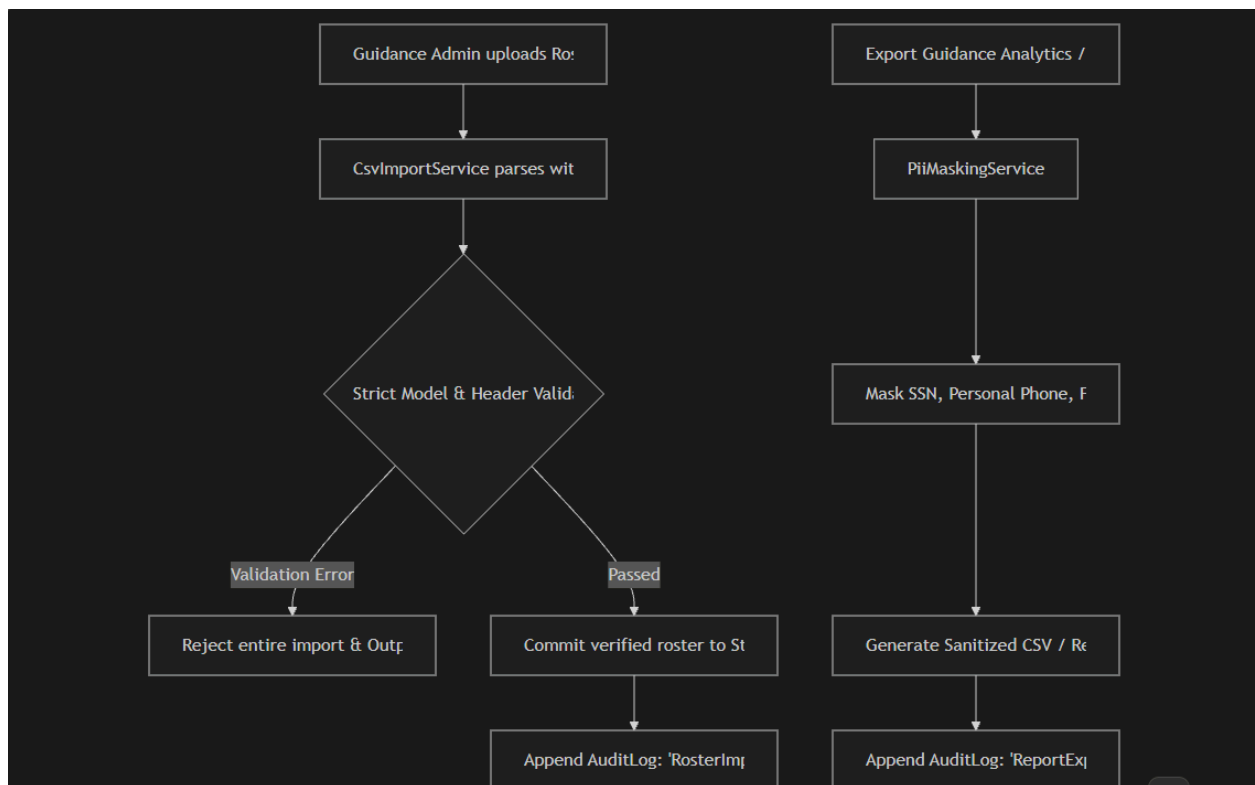
5.3 Inbound Referral Intake Workflow



5.4 Student Roster & Data Operations Flow



5.5 Guidance Request State Machine



State Definitions

- **Requested:** Case pending initial counselor review. Retains full submission + Student Claims.
- **ReferredIn:** Ingested from external department. Prioritized in queue. Retains StudentId, ReferralReason, OriginatingDeptCode, and OriginatingRequestId.
- **InReview:** Counselor actively conducting triage or counseling session. Retains local confidential counseling notes.
- **ReferredOut:** Request transferred to another campus department. Local case closed. Retains sanitized reference pointer and dispatch audit record.
- **Resolved:** Case closed with outcome summary. Retains resolution summary and immutable audit log entry.

6. Proposed Architecture / Technology

6.1 Architectural Principles & Boundaries

- **Canonical Reference Architecture:** GuidanceDepartmentMain serves as the single source of truth for shared services (Contracts/ServiceContracts.cs, AuthService.cs, NotificationService.cs).
- **Data Minimization & Privacy Safety Valve:** Counselor notes and mental health logs entered in triage.html must **never** leave GuidanceDepartmentMain. Only sanitized ReferralReason and StudentId are transferred across project boundaries.
- **Claims-Driven Identity:** StudentRequestService strictly extracts StudentId from JWT claims (HttpContext.User) to prevent spoofing.
- **Audit Immutability:** Lifecycle transitions write append-only records to AuditLogService.

6.2 Implementation Phases

Section to Update	Required Change
Shared Service Layer	Add ReferralService (or CrossAppTransportService) and InterServiceAuthHandler to the list of shared services copied from GuidanceDepartmentMain.
Contracts	Document ReferRequestDto and ServiceResult additions in Contracts/ServiceContracts.cs.
Dependency Registration	Update the Program.cs registration rules to include scoped registrations for inter-service referral handlers and HTTP client options.
Security & Production Boundaries	Add rules for Machine-to-Machine (M2M) authorization keys or shared JWT signing keys used when department applications call each other across ports.

7. Expected Benefits / Impact

1. **Seamless Cross-Department Collaboration:** Staff can forward student cases across departments without relying on manual, un-audited email chains.
2. **Uncompromised Data Privacy:** Sensitive counseling session notes remain isolated within GuidanceDepartmentMain while target departments receive only actionable referral details.
3. **Resilient Inter-Service Communication:** Integrated Polly circuit breakers and retries protect the system against target service downtime or restart cycles.

4. **Complete Auditability:** Every outbound and inbound referral leaves an immutable log entry in the respective department's audit store.
5. **Maintainable Codebase:** Using GuidanceDepartmentMain as the canonical source prevents code drift across all six independent projects.

8. Status & Action Plan

Deliverable	Type	Effort	Status
Phase A: Department Identity & Service Synchronization	Core Infrastructure	Medium	Not Approved
Phase B: Inter-Service Referral Protocol	New Inter-Service Workflow	Medium	Not Approved
Phase C: Department-Aware PII Masking & CSV Rules	Service Enhancement	Low–Medium	Not Approved
Phase D: Department Admin Surface (admin.html)	UI / Config Extension	Low–Medium	Not Approved
Design Variance Protocol: Custom Department Rules	Pre-Dev Process	N/A	Adopted (Revision 3)
Central AI Context Sync Plan: Context Alignment	Technical Documentation	N/A	Adopted (Revision 3)